

KEMAL ÖZKIRŞEHİRLİ

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I leverage AI/ML frameworks, generative/diffusion models, free-energy methods; atomistic molecular dynamics, Monte Carlo, and density-functional theory simulations; inference, robust optimization to design algorithms that transmute complexities into parsable, compilable, and consequently computable representations and structures. Thereafter, I extract the governing rules of these systems to synthesize high-level technical knowledge to drive critical decision-making.

Education

Massachusetts Institute of Technology:

Cambridge, MA, USA.

B.Sc.: GPA: 5.0/5.0 **Majors:** Computer Science and Engineering, Chemical Physics, Mathematics | **Minors:** Philosophy, Writing. **Expected May 2027.**

- **Research Interests:** Biomolecular Simulation, Free-Energy Methods, CADD, Statistical Mechanics-Renormalization Group Theory, Chemical Diffusion, High-Performance Computing, Generative Models, Scientific Workflow and DNA-RNA Engineering, Agentic AI, Inference.
- **Coursework:** Deep Learning | Algorithm Engineering | Natural Language Processing | AI-Decision Making in Medicine: from Disease to Therapy | Advanced Algorithm Design-Analysis | Theory of Computation | Stochastic Processes | Sublinear Time Algorithms | Computer Structures and Architectures | Python and C/C++/Assembly Programming Labs | Probability Theory and Random Variables | Linear Algebra | Advanced Organic Chemistry | Discrete Mathematics.
- **Humanitas Core:** Theory of Models | Rhetoric | Advanced Essay, Poetry | Philosophy of Language, Religion, Politics | Metaphysics | Memoir.
- **Activities:** AI@MIT, Schwarzman College of Computing Advisor, BioMakers, MIT Student Council Representative, ClubChem, MIT Gala Model/Designer.

Research, Industrial, and Professional Experiences

PRISM AI Safety Fellow, Protein Foundation Model Red-Teaming and Evaluation: *PRISM, W2D2, Siemens*

May 2026 – Present.

- Selected from +800 applicants for PRISM 2026 to develop adversarial evaluation methods for protein foundation models, testing when biologically plausible outputs fail sequence-structure-function constraints, uncertainty calibration, and protein-design reliability checks.
- Creating methods to test whether or not a protein design model will be reliable during high-stakes workflow applications by considering several areas, including biological plausibility, structural consistency, developability, uncertainty calibration, mechanisms that the models claim to exist, and failure behaviors of these models when designing proteins.
- Building a safety-bounded benchmark and failure taxonomies for next-generation protein model designs using previously developed work in EVEdesign, A*STAR V2M, antibody/protein models, CADD, and uncertainty-aware candidate ranking to determine robustness and safe science use.

Summer UROP Scholar for ChromoGen-Engine-V2, Prof. Zhang: *MIT Chemistry, Computational Systems Biology*

May 2026 – Present.

- Developing ChromoGen-Engine V2, a conditional diffusion and evaluation pipeline that merges ChromoGen-style single-cell 3D chromatin generation with small-molecule chemical perturbation data and same-cell multi-omic regulatory readouts, generating chromatin distance-map ensembles conditioned on genomic sequence, perturbation labels, and/or scATAC-derived regulatory state features.
- Building data engine integrating Plate-C / Easy Dip-C small-molecule treatment structure data with CHARM-style same-cell Hi-C, RNA-seq, ATAC-seq, histone-modification data, using leakage-aware splits across treatment identity, dose/time metadata, cell type/state, genomic window, batch/source.
- Designing a structure-grounded benchmark to test whether chemical perturbation or regulatory-state conditioning improves generated 3D chromatin ensembles beyond sequence-only, population-ATAC, pseudobulk/metacell, shuffled-label, and ablated baselines, with metrics including held-out distance/contact agreement, P(s) decay, condition separation, ensemble coverage, distributional calibration, failure-mode analysis, and interpretable attribution of sequence/regulatory features linked to predicted structural shifts.

Summer AI/ML Researcher for Autoimmune Target Discovery: *Experimental Drug Development Centre, A*STAR*

May 2026 – Present.

- Architecting V2M-Engine with Dr. Sun and Dr. Cai, a calibrated genetics-to-mechanism target-discovery framework that maps autoimmune GWAS/fine-mapped risk variants onto regulatory programs, causal immune cell types/states, and targetable disease mechanisms using single-cell foundation models, QTL/colocalization evidence, and known therapeutic targets as embedded validation controls.
- Building a Target-Mechanism Retrodiction Benchmark that pressure-tests whether scGPT/Geneformer-derived cell-state embeddings recover known immune-disease mechanisms beyond classical enrichment, marker-gene, pathway, and eQTL/colocalization baselines, with ablation studies, uncertainty quantification, calibration error analysis, and leakage-controlled target-family holdouts.
- Developing modular sequence-to-cell-state-to-targetability engine coupling Evo 2/EVEE-style variant-effect scoring, perturbation/Perturb-seq validation, counterfactual immune-state transport, structure/affinity-aware target prioritization to rank paths that are biologically interpretable/drug-discovery-actionable.

Student Principal Investigator, Özkırşehirli Group and TBXT Small-Molecule Hackathon: *MIT, Pillar VC, onepot*

April 2026 – Present.

- Leading 11-person chordoma-focused TBXT/brachyury computational hit-identification sprint targeting PDB 6F59 chain A / TBXT G177D site F; compressed 2,274 prior-art compounds + 737 raw analogs → 503 filtered analogs + 30,000 BRICS recombinations → 67 novel QSAR-pass proposals into a 570-compound novelty-filtered pool using site-F/A/G grids, Tanimoto novelty control, and sourceability-aware generation.
- Integrated multi-orthogonal AI/CADD stack: Vina ensemble docking, GNINA CNN pose/pKd scoring, Vina-trap detection, RF/XGBoost TBXT QSAR, Boltz-2 co-folding, MMGBSA/FEP scaffolding, T-box paralog selectivity, Rowan IC50/affinity analysis, RDKit descriptors/BRICS, onepot/muni catalog checks, and Bash/HPC automation; trained QSAR on 650 RDKit-valid SPR-derived compounds from 14 decrypted XLXS / 15 campaigns / 1,620 Kd fits, achieving Spearman $\rho \approx 0.49$ and MAE ≈ 0.5 pKd; GNINA screened 569/570 and flagged 40 Tier-A, 51 Tier-B, and 73 Vina-trap candidates.
- Prioritized final computational submission funnel 570 → 137 strict-pass → 24 submission-ready → 4 judge-facing site-F picks under 100% onepot exact match, non-covalent chemistry, PAINS/forbidden-motif exclusion, MW ≤ 600 , LogP ≤ 6 , HBD ≤ 6 , HBA ≤ 12 , lead-like, ESOL/logS, Tanimoto <0.85 , cost/chemistry/supplier-risk, and 16-paralog selectivity gates; final 4 showed Boltz Kd 3.2–8.8 μM , Jack/SCC agreement 1.01–1.34 $\times$, GNINA Vina -5.01 to -6.19 / pKd 3.94–4.69, \$875 total sourcing cost, and Rowan computational readouts covering 10/19 assigned/co-assigned completed candidates, including all 4 final picks with IC50-style predictions 1.82–6.11 μM .

LLM and Statistical Mechanics/Theoretical Chemistry Researcher, Prof. David Reichman: *Columbia University, NY*

Dec 2025 – Present.

- Developed ChemAgent-QSM: an LLM-guided agentic framework converting natural-language chemistry prompts into validated quantum chemical pipelines using PySCF for electronic-structure analysis of drug-like molecules.
- Statistically generates auditable Python workflows for time correlation functions (TCFs), local order metric, relaxation time scale, mobility field, and structure/dynamics coupling of disordered condensed phase, optimized geometries, electronic-structure descriptors, and vibrational spectra via ab initio DFT, automated end-to-end QM workflows bridging electronic-structure theory and drug discovery.
- Developed LLM-trajectory features and machine learning analysis workflows that can be used to determine dynamical heterogeneity in glass-forming and other systems with statistical mechanics baseline methods: mean squared displacement, SISFs, and MSCOPE.

Protein Design Algorithm Engineer, EVEdesign Collaborative Project, Prof. Marks: *Harvard Medical School*

April 2026 – Present.

- Working on EVEdesign's open-source, method-independent protein-design platform as part of a 21-person, 18-institution, 8-country collaboration, supporting composable multi-objective antibody/protein design workflows across evolutionary, structural, developability, and experimental-feedback constraints.
- Developing algorithmic and engineering components for uncertainty-aware candidate selection, sequence-structure-function objective integration into scoring functions using protein-language-model/evolutionary priors, and agent-based tool orchestration for scalable, lab-in-the-loop biosequence-design pipelines.

Head of Chemistry Research-Implementation, Backend Software Engineer for LangGraph AI Agent Pipelines: *Pedal AI* **Jul–Dec 2025.**

- Designed a high-performance backend architecture for LangGraph AI agents using Python, implemented FastAPIs for connections between AI agents and molecular prediction engines, optimized control flow to scale model interactions and data processing/evaluation, spanning 3 core backend layers and 4,000+ lines of code, reducing average prediction latency to <50ms while supporting 15,000+ daily API requests and processing 50+ GB of chemical datasets.
- Built chemist- and researcher-friendly UI/UX for Pedal AI's Agentic Chemistry Platform to include Coley Lab retrosynthesis models with ASKCOS-style route search capabilities as part of inspectable retrosynthesis trees, reaction-level provenance, and decision-ready synthesis workflows, mapping +8-step synthetic pathways and successfully evaluating over 100,000 complex molecular predictions with 99.9% system uptime.

Project Co-Lead, Deep Reinforcement Learning for Antibody-Antigen Interactions: *AI@MIT / AIM Labs, MA, USA* **Jan 2026 – Present.**

- Co-led a team of six researchers, developed a structure-informed cross-attention transformer and ESM-2 pipeline that maps protein/pathogen sequences to candidate antibodies, curated datasets from OAS, SABDab, and IEDB.
- Reduced training time by ~40% after proximal policy optimization-based reinforcement learning against AlphaFold-Multimer/IGFold structural metrics using GPU-distributed PyTorch DDP on MIT SuperCloud Supercomputer.

AI/ML for Statistical Mechanics Simulations Researcher and Teaching Fellow, Prof. A. Nihat Berker: *MIT Physics* **Jan 2022 – Present.**

- Invited to conduct research after ranking within top 1% in Prof. Berker's graduate-level "8.334: Phase Transitions and Renormalization-Group Theory" in high school. Building a machine learning-augmented Kadanoff renormalization group framework using PyTorch with a graph neural network (GNN)-based spin classifier to automatically find fixed points and map out thermodynamic phase boundaries in spin-1/2 antiferromagnetic systems.
- Mentored 200+ students regarding college applications/decisions and career paths while TA'ing three condensed courses Augmented Chemistry, Classical/Quantum Mechanics (two-week, daily exams and homework), graded problem sets and exams approximately 6-7 hours per day for one month in total.

Kupcinet-Getz Scholar in Computational Biochemical Diffusion, Prof. Semenov: *Weizmann Institute of Science, Israel* **Jun–Aug 2025.**

- Selected as one of 22 scholars worldwide (< 10%), developed PINN- and neural ODE-enhanced stochastic simulation algorithms of nonequilibrium enzyme-reaction/diffusion networks, developed custom Euler-Maruyama, Runge-Kutta, and Gillespie SSA solvers for biochemical kinetics and spatial heterogeneity.
- Constructed GNN encoder and transformer sequence models with UMAP-driven clustering to extract latent thermal structure from high-dimensional trajectory datasets, constructed PyTorch classifiers to automatically identify bifurcations and nonlinear dynamics from high-dimensional trajectories.

Chemistry and AI Consultant and Scientific Data and Evaluation Specialist: *Sepal AI, Mercor* **Sept 2024 – Feb 2025.**

- Developed model-training, model-testing, and model-evaluation data for front-line LLM workflow applications that can convert graduate-level problems involving retrosynthesis, reaction mechanisms, and method selection to a format that can be evaluated against a known correct solution. Developed grading rubric and chemical validity constraints for each task type.
- Developed instructional data, prompt engineering techniques, and workflows used for LLM fine-tuning and response generation based on curating instructional data, analyzing errors, decomposing prompts, and annotating failure modes for advanced chemistry reasoning; converted this instructional data into structured formats including instruction/response, evaluation/grading, and feedback formats for both model training and benchmarking.

Foundation Research Scholar in Computational Organic Chemistry: *Lumiere Research Inclusion Foundation* **August 2023 – February 2024.**

- Engineered an automated multi-scale simulation pipeline linking quantum mechanical calculations (DFT) to stochastic kinetic Monte Carlo (kMC) models; screened 20+ transition states and 500+ reaction trajectories across 5 solvent pathways to achieve a 22% predicted yield optimization for a cytotoxic target.
- Secured a highly selective full research scholarship (top 5% of applicants) to independently drive this computational project, collaborating 1:1 with a Brown University PhD mentor to analyze reaction dynamics and compile a rigorous mechanistic manuscript.

Advanced Organic Synthesis Researcher, Prof. Stephen L. Buchwald: *MIT Chemistry, MA, USA* **Dec 2025 – May 2026.**

- Augmented the development of new CuH-catalyst to perform selectively asymmetric methylation and alkylation of vinyl boronate esters, determined a structural relationship between the catalyst's structure and its activity through ¹H-NMR and chiral-HPLC analyses, expanded the number of usable substrates, scaled-up synthesis from laboratory scale (0.1 mmol) to multi-gram scales without changing physical or chemical properties.

Teaching Fellow and Grader for 6.1200[J]: Discrete Mathematics/Mathematics for Computer Science *MIT EECS* **Jan 2026 – Present.**

- Selected to support teaching staff after receiving an A+, ranked in the top 1% of students enrolled in MIT EECS's largest foundational theoretical computer science course, and grading for Spring 2026 the proof-intensive assignments that were focused on topics such as logic, graph theory, recurrence relations, asymptotic analysis, cryptography.
- Assisting in guiding a 250+ student cohort per semester via recitation sessions, office hour sessions, and grading for midterms and finals.

Honors, Awards, and Recognition

- **Medalist in 56th International Chemistry Olympiad**, representing Türkiye; ranked 105th internationally among 327 participants.
- **Medalist in 58th International Mendeleev Chemistry Olympiad**, representing Türkiye; ranked 73rd internationally among 151 participants.
- **Silver Medal in TÜBİTAK National Chemistry Olympiad**, ranked 3rd nationally among 62 finalists and 2500+ students to represent Türkiye.
- **Gold Medal in Istanbul Science Olympiads 2022 in Chemistry**, awarded for ranking first in Istanbul out of 50 finalists.
- **Columbia Science Research Fellow Named Scholar**, awarded full-ride early with Likely Letter in recognition of scientific accomplishments and talent.
- **World Science Scholar 2023 Cohort**, selected to 52 scholars from 22 countries for exceptional mathematical talent and fourth Turkish scholar in history.
- **AIME Qualification and Acknowledgement in AMC Hall of Fame**, achieved 7th place in Turkey and top 5% grade out of 300,000+ participants.
- **Selected to attend AlgoTrade Hackathon 2026** with free travel and accommodation from 1,000+ applicants across 51 countries and 148 universities.
- **Transferred to MIT from Columbia**, one of 22 from 764 with 2.5% acceptance rate, deemed by MIT officers to be "the hardest path to get into MIT."
- **CHEM13 Certificate of Distinction**, University of Waterloo International Exam, ranked 2nd among 1,551 participants.
- **Certificate of Distinction**, Galois International Mathematics Contest, ranked 7th among 3071 international students, 20th globally among 6595.
- **Book Prize**, Sir Isaac Newton International Physics Contest, University of Waterloo, ranked in the top 50 globally among 2000+.

Personal

- **Computational Skillset:** Python, C/C++, CUDA, HPC/GPU Programming, Linux, Bash, Git, SLURM/PBS/qsub, Conda, Distributed Training, NumPy, PyTorch, MD, OpenMM, OpenFF, OpenFE, MDTraj, ParmEd, PDBFixer, RDKit, Computer-Aided Drug Discovery (CADD), SMILES/SDF/PDB/PDBQT, Molecular Docking & Virtual Screening, AutoDock Vina, GNINA CNN Scoring, Boltz-2 Protein–Ligand Co-Folding, MMGBSA/FEP/Free-Energy Methods, QSAR (RF/XGBoost), ADMET/Rowan, Morgan Fingerprints, Tanimoto/PAINS/ESOL Filtering, BRICS/Generative Chemistry, onepot/muni Catalog Screening, PySCF, PyMOL, SciML, Transformers, LLM Embeddings, Sequence Modeling, RL: PPO, GNNs, SQL, Java, MATLAB, LaTeX.
- **Languages:** English (Native), Turkish (Native), German (Professional), Spanish (Beginner), Hebrew (Beginner).
- **Interests:** Philosophy of Science, Epistemology, Logic, Metaphysics, Psychology, Language, History, Aesthetics, Ethics, Politics, Power, Biopolitics, Socio-cultural Anthropology, Foucauldian Studies, Love; Judaism, Islam, Critical Theories: Queer, Postmodern, Psychoanalytic, Post-structuralist, Race, Feminist.
- **Hobbies:** Competitive Volleyball, Chess, Poker, Personal/Political/Academic Essay Editing/Writing, Figure Skating, Meditative and Labyrinth Walking.